Exploring the Vulnerabilities and Advantages of SWIPE or Pattern authentication in Virtual Reality (VR)

Ilesanmi Olade
University of Liverpool
Liverpool, UK
Ilesanmi.Olade@liverpool.ac.uk

Hai-Ning Liang
Xi'an Jiaotong-Liverpool University
Suzhou, China
Haining.Liang@xjtlu.edu.cn

Charles Fleming
University of Mississippi
Mississippi, USA
fleming@olemiss.edu

Christopher Champion
Xi'an Jiaotong-Liverpool University
Suzhou, China
c.champion17@student.xjtlu.edu.cn

ABSTRACT

Virtual reality applications are carving out a new niche within the entertainment and business user-sphere, therefore reliable security and usability are essential to achieving consumer confidence. In this paper, we are exploring (1) the suitability of porting the popular SWIPE mobile device authentication system for use within virtual reality (VR) by observing the advantages and vulnerabilities. (2) The effects of the interaction devices such as the hand-held-controller (HHC), the LeapMotion sensor, EyeTracker and the head-mounted-display (HMD). Our study is in three-folds, a web study (N=219) to collect and analyze possible SWIPE password patterns, then a mobile device study (N=15) and a VR study (N = 15) to evaluate the speed,login errors, usability of the SWIPE authentication system in both environment for comparison. We are also interested in the effectiveness of shoulder-surfing within VR as it is known to be a weakness in mobile devices.

CCS CONCEPTS

• **Security and privacy → Usability in security and privacy**; *Social aspects of security and privacy*; • **Human-centered computing → Virtual reality**.

KEYWORDS

Security; Virtual reality; PIN; Swipe; Pattern; Authentication; Mobile devices

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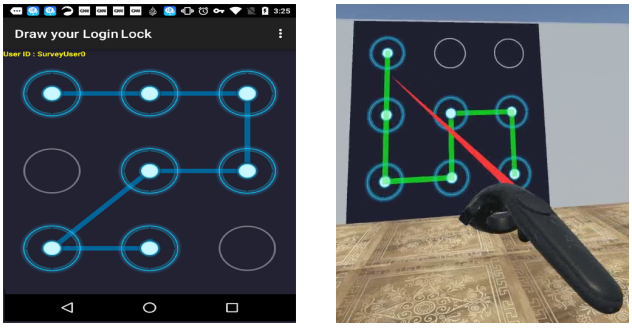
1 INTRODUCTION

Virtual Reality (VR) has quickly gained a large user base as the technology becomes cheaper, mainstream and more applications exist for this new environment. We observed its usage in education[1], medical [2], shopping[3], advertisement[4] and manufacturing[5] sectors. Some innovators and concept developers in the industry see VR as an evolution of the popular mobile device market and rely on existing mobile device consumers in that sector to promote and sustain the growth of this new trend[6]. Global online retailers and e-commerce platforms, such as Amazon and Alibaba have introduced virtual malls where VR enabled clients can walk through the shop and virtually interact with the products on sale. Virtual reality is here to stay and as the application of virtual reality continues to expand, the issue of user security and the lack of a standard authentication becomes increasingly apparent [7]. We believe that VR and mobile devices, especially smartphones and tablets are intrinsically linked in terms of usage, while other researchers[6, 8] are exploring the development of new authentication technologies for the VR environment such as biometrics [9, 10] and kinesiology [11, 12], we are looking at porting an existing mobile device authentication system and observing how they perform in this new environment. This has the advantage of pre-existing popularity and familiarity. We are vividly aware of the security challenges plaguing the mobile device authentication systems as large body of research [13, 14] exist in that field and we wish to explore the possible fallacies and advantages that a VR version of these authentications may have considering the fact that VR users are a large subset of existing mobile device users. Our research focuses on examining the performance and usability factors of the popular mobile device authentication system known as SWIPE lock within the VR environment. This system was made popular by the Android smartphone market and it is now practically the defacto authentication system on every mobile device.

The SWIPE authentication system is a grid-like display of nodes (see Figure 1) whose password pattern is selected by continuous finger or pointer movement across the screen to connect the password nodes. SWIPE has been plagued by numerous usage and security issues [15, 16]in the mobile device usage ecosystem, and one of the most prominent security weakness is Shoulder-surfing. We argue



(a) SmartPhone SWIPE (b) VR SWIPE

Figure 1: Prominent mobile device authentication systems: The SWIPE authentication system is popular with (a) mobile devices that have GUI touch screen based systems. (b) Shows the VR SWIPE ported to the Virtual reality environment.

that the VR version of SWIPE would be highly resistant to shoulder-surfing and will also perform comparatively well in other usage aspects such as speed, error-recovery, and high entropy.

2 RELATED WORK

Consumer electronics have made security an important factor in the usage and design of user devices, therefore authentication exists in many implementations with three core types of authentication categories which are namely *What you know*, *What you are* and *What you have*. Virtual Reality as an emerging consumer product has a limited number of studies on security, therefore we shall examine related works on authentication in mobile devices and personal computers due to their important derivative relevancy to the subject matter. Additionally, we shall consider recently available studies on VR user security, but keeping the scope limited to *non-biometric* or *non-token* based solutions.

2.1 Mobile Devices

Text Based Passwords Since the 1960s, text-based passwords [17] have dominated devices that require keyboards and keypads for standard input. This usage transitioned to personal mobile devices and was graphically represented on touchscreen mobile devices [14]. PIN authentication and TEXT authentication are equally popular and are both plagued by weaknesses and usability issues that have been extensively explored over the years[18]. Melicher et al. [19] and Bonneau et al.[20] explored theoretical and practical password spaces. Other studies based on exposed large databases[21, 22] explored password security issues that were a result of user behaviour[23], lack of motivation[24], device interface factors[25] and socio-cultural influences [19].

Graphical Passwords Graphical passwords provide a promising alternative to traditional alphanumeric passwords. it is an authentication system that is presented to the user via a touch-screen based GUI that enables easy interaction [26]. They are attractive and intuitive since people usually remember shapes and images better than random words or text. In recent years, various studies have categorized graphical authentication methods into 3 main categories, which are : *Recall* [27], *Recognition*[28], *Cued-recall*[15, 29]. The Android SWIPE graphical authentication system is based on the Recall category.

2.2 Virtual Reality

In the area of Virtual Reality authentication, various research has been mainly focused on the *What you are* aspect of authentication paradigm. Biometric authentication research [9] is very popular for VR because virtual reality inherently exposes various biometric properties of the virtual reality user and these studies are gaining popularity. However, there is growing hesitation to biometric authentication for privacy and data safety reasons. Our paper focuses only on the well known *knowledge-based authentication* in virtual reality systems.

A study by Yada et. al [30] using PIN authentication in augmented reality systems found encouraging results even though the password entry time of this study was five times higher than using PIN on a standard mobile phone. The researcher suggested that visual cues and the reaction time of the Google Glass device was a contributing factor. A recent study by George et al.[31] evaluated the security and usability of SWIPE and PIN within the VR environment. The study involved 25 participants, they found that usability was comparable in performance to the mobile version of PIN and SWIPE. Although their study was analytically sound, it did not evaluate the shoulder-surfing threat and lacked a direct comparison to an actual mobile device study involving SWIPE and PIN.

3 METHODOLOGY

3.1 Participant recruitment and ethical concerns

Because our study utilizes human subjects, it was reviewed and approved by our University Research Ethics committee. One concern raised was whether or not participants would inadvertently reveal their personal passwords. To prevent this, a text was added in the instructions of each experiment specifically warning participants not to use any current or past personal password as part of their responses. Participants for all experiments were volunteers, recruited via a university-wide e-mail. Participants were primarily students, but also included some staff members. All participants were daily mobile device users. Participants in the web survey portion of our study were not compensated, due to the number of participants and the relatively short duration of their participation. Volunteers for the longer-term memorability and usability studies were compensated with coffee shop gift cards worth between 5 and 30 USD, depending on the duration of participation.

3.2 Experimental Design

Our goal was to compare the mobile device and VR SWIPE systems and their interactions with other independent variables. To do this, we followed a within-participants design.

3.2.1 ***Study Outline.*** We utilized three studies for data collection in an attempt to expose and collect the data we need for our planned evaluation and comparison. The studies conducted were a web-based study, a mobile study and a virtual reality study.

Web Study: An Internet based study, where a large number of participants were allowed to create a series of SWIPE passwords via a web-based version of the SWIPE system (see Figure 2a). The collected password was used for analysis (see Section 3.3).

Mobile Device Study: A mobile device study, using Android based mobile phones (see Figure 1a). Participants were tasked with creating and using the SWIPE password over a period of time. Data collection and real-time participant surveys were done via the custom built android SWIPE authentication system (see Section 3.4).

Virtual Reality Study: This study involved creating a virtual reality version of the SWIPE authentication system (see Figure 1b). Participants interacted, using 4 different interaction methods, and created passwords while various metrics were tracked and logged for analysis (see Section 3.4). We also made video recording of these sessions from 3 different points of views.

3.2.2 ***User Interaction Methods.*** Participants used five different methods to interact with the SWIPE authentication system. The mobile device interaction is solely via the touch-screen, whereas the virtual reality environment SWIPE authentication system entails four interaction methods:

- **Hand-Held-Controller (HHC)**
- **Head-Mounted-Display (HMD)**
- **LeapMotion**
- **Eye-Tracking**

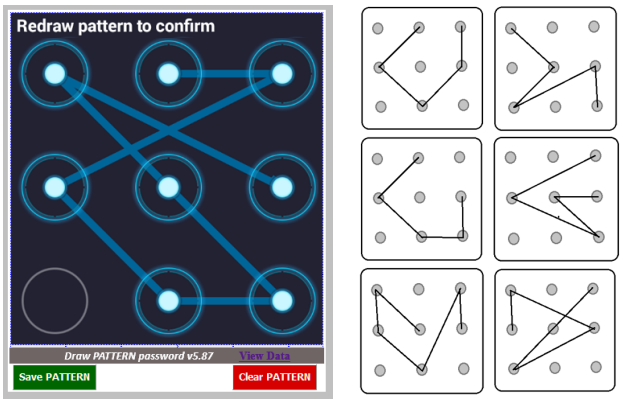
3.2.3 Study Hardware.

- **Touchscreen Mobile:** Mobile devices are available in numerous dimensions. We performed our study with a 5.2" LG Nexus 5X phone [32].
- **Virtual Reality Hardware:** The Virtual Reality system we used for our study is the HTC VIVE hardware [33] with standard handheld controllers. Third-party add-on hardware such as aGlass [34] for Eye Tracking and LeapMotion Hand tracking [35] were also used to provide additional tracking function and create a better realistic immersing VR experience.

3.3 Web-based Study

We presented the participants with a 3x3 SWIPE web-interface (see Figure 2a) and requested that they create 10 unique passwords that have more than 3 nodes. Although web-based experiments are harder to control than laboratory or supervised field studies, this channel of data collection meets our requirements, offers numerous advantages and allowed us to collect large amounts of data from our participants at various locations. Each participant was tracked via their device Internet Protocol (IP) address and their chosen SWIPE patterns were logged in a database.

3.3.1 Goals. Our primary goal was to quantify the effect of a participant's choice on the security of passwords chosen. Every authentication scheme has an entropy and the strength of such entropy is determined by the probability distribution associated with the password space. Ideally, this distribution is approximately uniform. We selected from the provided passwords, six SWIPE patterns that we decided were well distributed across the password nodes (see Figure 2b) to be used in the Mobile Device and Virtual Reality study.



(a) SWIPE Web Study Interface **(b) 6 SWIPE Passwords**

Figure 2: Web Study: The SWIPE Web Study presented a Web version of the SWIPE interface (a) to participants who were asked to create random passwords. From the numerous passwords we selected 6 passwords that were complex and uncommon (b).

3.3.2 Participants. We created awareness about the Web study via social media and the university mailing list. A tutorial video was made available on the website so participants could watch and learn how to create SWIPE password patterns. We had no input in selecting the participants, we did not collect any demographic information about these participants, but we ensured their IP was unique.

3.4 Mobile Device & Virtual Reality (VR) Study

We developed a mobile device version (see figure. 1a) of the SWIPE interface that was used during our web-based study (see figure. 2a). This mobile device authentication system will be our baseline or control for this study due to the popularity and large body of research literature [13, 15, 36] about its performance. It is important to note that due to the peculiarity of this VR environment (see Figure 1b) we explored 4 different methods of user interactions (see Section 3.2.2). Each participant used 5 different interaction techniques (see Section 3.2.2) and completed 3 SWIPE login trials per interaction technique. During the VR session, the participants physical Login movements or actions were recorded with a video camera. We captured their actions (see Figure 3) from 3 different points of views (Front, Top and Side). These recordings will be used to analyzed the level of resistance that virtual reality (VR) has against Shoulder-surfing. Lastly, we provided a paper based survey questionnaire to collect qualitative feedback from our participants in a uniform and consistent way.

Our goal during the study, which involved participants, was to collect both qualitative and quantitative data which would provide insight into our participant's perception of the likeability, usability, memorability and login speed of both types of SWIPE authentication system (see Figure 1). During the testing, a participant's activities such as touches, password tokens, strokes, pauses, timings, aborts and errors were logged for further analysis.

We recruited 15 participants (7 females) from a local university and 51% of the participants were between the ages of 17 to 27 and

all our participants were right-handed. All were active users of mobile phones and 60% had experience with Virtual Reality devices. 70% of them used a phone with a fingerprint sensor, while 10% used the PIN, 20% used Swipe.

Our first step was to inform the participants about the confidentiality of their supplied information and to explain the purpose of the project and the tasks they would need to do. We required that everyone start with the mobile device environment, then conclude with the virtual reality environment. We provided a five minute training video to each participant that depicts the usage of SWIPE in the mobile device and virtual reality environment. We calibrated each participant's eyes for the aGlass EyeTracker software and created a profile for each person and also explained the usage constraints of the LeapMotion device, then we allowed participants to practice with the Virtual Reality equipment and alerted them to the possibility of motion sickness while in the virtual reality environment.

Week 1 (First Phase): We allowed each participant to choose a SWIPE password from our supplied list (See Figure 2). During the experiment, the participants are required to enter this selected password 3 times during the session. If the participant entered the wrong password, the application alerted them to enter the correct password again. The experiment finished with a Likert questionnaire that collected qualitative data about the participants' perceived *usability, error-handling, security and likeability.*The average time the participants used to complete the mobile device experiment was 2 minutes while the VR experiments took about 14 minutes.

Week 6 (Second Phase): In the second phase, i.e 30 days after the first phase, we explored the effects of shoulder-surfing within the VR environment. In this session we used 12 participants as "Attackers", we selected 2 persons each who were not involved previously with the study as *Type A and B* attackers, and used 8 persons from week 1 (see section 3.4) of our project who we confirmed were unable to remember their week 1 Login passwords as *Type C* attackers. *Type A* attackers were not allowed to view the videos, while *Type B* attackers viewed the videos and wrote down 3 possible passwords based on their visual evaluation of the participant's actions. *Type C* attackers viewed only their own videos. The outcome of this session is analyzed in Section 5.1.

4 DATA COLLECTION & MEASUREMENT

Pre-Login Delay time Pre-login delay is the elapsed time between when the participant indicated that they were ready to start unlocking the device and the actual time they entered the password. This data provides a view into evaluating the memorability and usability of the system. Studies by Stobert et al. [37, 38] defined a direct relationship between memorability and pre-login delay time. We analyze this data to quantify the level of memorability and usability experienced by the participants.

Login Speed The time period used to complete each trial of the login process for an interaction technique was recorded. This measurement only recorded successful trials; failed trials were recorded as singular failure events. Login speed was tracked from the moment a participant starts password token entry until the entry was completed successfully.

Error Rate The error rate was measured as a percentage of failed login attempts to the total number of attempts required to complete the technique's session. The number of failed login attempts during a trial did not affect the number of trials that constituted a complete session.

Shoulder-surfing Attack Evaluation Shoulder-surfing is a known weakness of SWIPE on mobile devices[15, 30, 36, 39]. We setup three high resolution cameras to record the physical actions of the participants. These cameras captured the *Front, Top and Side* views of participants while ensuring that their entire body was visible as the performed both the mobile device and VR sessions (see figure 3). The video feed was multiplexed into a single video allowing the 'attackers' to observe the participants simultaneously from all views.

Figure 3: Shoulder-surfing evaluation was performed by recording the participants Login actions via 3 video cameras and having future 'attackers' view these recordings in an attempt to guess the passwords.

Subjective Data We collected *pre-test*, *in-test* and *post-test* surveys via an electronic questionnaire. The questions focused on ease of use, perception of speed, the likelihood of adoption, error recovery, and interface usability. We implemented the questionnaire in electronic form and used 5-point Likert questions for some aspects of the questionnaire.

5 RESULTS

We present the results of our study in both *quantitative* and *qualitative* formats. This analysis is based on the raw numerical data and subjective data we collected from the participants.

5.1 Quantitative Results

Login Speed The Login speed data from the study indicates that *MD:Touchscreen* login speed was faster than any of the other VR interaction technique login speed. The mean values of the login speed of each interaction technique are shown in figure 4. The one-way ANOVA test ($F(4,461) = 120.19$, $p < .031$) indicates statistical differences between the interaction techniques. A Tukey post hoc test revealed that *MD:Touchscreen* login speed (913.04 ± 172.53 ms,

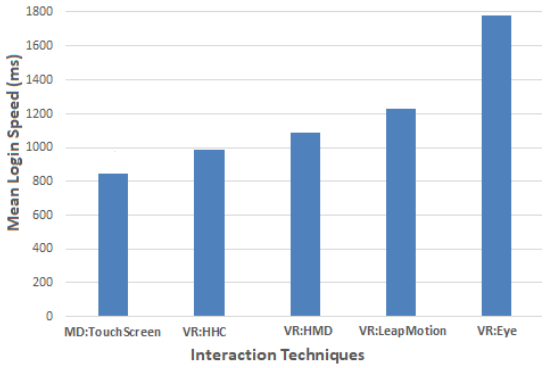


Figure 4: **Login Speed: Mean Login speed based on interaction techniques. The mobile device Touchscreen has the lowest login time.**

p < .001) was significantly faster than all VR interaction techniques, and there was also statistically significant difference between the *VR:HHC* and the *VR:Eye* but there was no statistically significant difference between the *VR:HHC* and *VR:HMD* interaction techniques (960 ms, p = .965) or the *VR:LeapMotion*.

Pre-Login Delay Time Our participants experience a time delay between when the trial started and when an initial action or interaction was made. This pre-login delay time gives an indication of familiarity or ease of use of these interaction techniques. Our analysis indicates that the *MD:Touchscreen* has the lowest Pre-Login Delay time and the VR:Eye interaction technique has the highest time (see Figure 5). This is understandable since the virtual reality environment was novel to most participants.

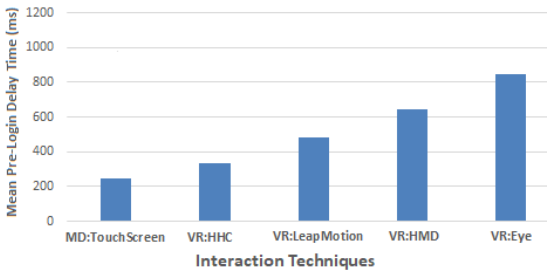


Figure 5: **PreLogin Delay time: The PreLogin Delay Time is an indication of ease of use or familiarity of the interaction techniques**

Shoulder-surfing Attack results Our 12 attackers evaluated the videos and provided 3 passwords each for the 15 sessions. The 3 passwords could be of the *3-Node match*, *4-Node match* or *full match*. We observed that 97% of the sessions used a 5-Node password. The results indicates that the VR environment is highly resistant to shoulder-surfing with the *Type C* attacker achieving a *Full Match* of 36% while *Type B* attacker and *Type A* attackers have a 20% and 3% success rate respectively. We allowed the Type B and C attackers to view the video recording of the mobile sessions and the results were 100% Full Match for all sessions, this supports the fact that mobile devices have little or no resistance to well implemented forms of shoulder-surfing. (see Figure 6).

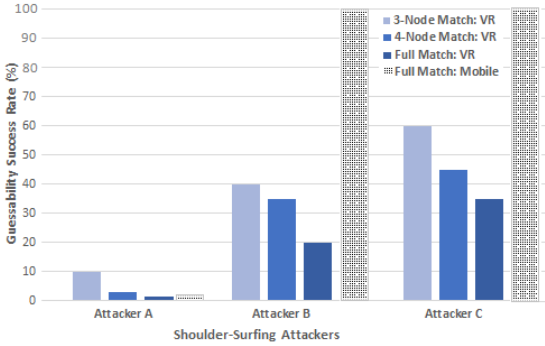


Figure 6: **Shoulder-Surfing Attack: Results from the Shoulder-surfing attack process. We defined the attackers and provided different knowledge and access to recorded videos, then allowed these attackers to guess the SWIPE passwords that they observed.**

Error Rate

Error rates classified by interaction techniques is an indication of the usability level of the interaction techniques. As mentioned earlier, most participants are new to the VR environment, but we notice that the error rates of the *VR:Eye* and *VR:HMD* were very high. Some system errors due to the *VR:Eye* hardware malfunction were accounted and adjusted for. A two-way ANOVA test was conducted to examine the error rate for each technique. There was no significant effect of interaction by these independent variables on the error rate. Furthermore, analysis showed that error rate was lowest for *MD:Touchscreen* and there was significant difference in the error rate of the *MD:Touchscreen* technique across all interaction techniques (p = .001).

5.2 Qualitative Results

The results are based on a 5-point Likert scale questionnaire and subsequent user rankings of the five interaction techniques. Each participant prior to the experiment answered a pre-test survey which we used to obtain demographics, personal information, and mobile device/VR usage experience. The Likert scaled questions were answered after the trial of each technique to collect their subjective preferences. The data we collected was analyzed using the Friedman test and we performed post hoc analysis with Wilcoxon signed-rank test with Bonferroni correction (p= 0.05/3 = 0.017) of those that are statistically significant. In the questionnaire, we probed aspects of the users‘ experience with the five login interaction techniques.

Speed Our participants‘ experience with each interaction technique's speed shows there was a statistically significant perceived difference in speed depending on technique ($\chi^{2(2)}$ = 18.321, p < 0.001) (see Figure.7). Post hoc analysis indicated that there were no significant differences between *VR:HHC* and *VR:HMD* trials (Z = -2.101, p = 0.036) or between *VR:HMD* and *VR:LeapMotion* trials (Z = -1.560, p = 0.119). However, there was a significant difference in speed between *MD:Touchscreen* and *VR:HHC* trials (Z = -3.573, p < 0.001).

Likeability Post hoc analysis indicated that there was no significant difference in how well participants liked the techniques.

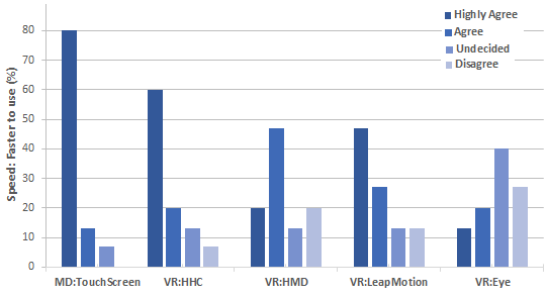


Figure 7: Perceived Login Speed: A comparison of the users' perceived speed for each interaction technique.

Usability There was a significant difference in perceived ease of use of technique ($\chi^{2(2)}$ = 14.22, p = 0.001). Post hoc analysis indicated that there were no significant differences between the *MD:Touchscreen* and *VR:HHC* (Z = -1.672, p = 0.94) or between the *VR:HHC* and *VR:LeapMotion* (Z = -1.628, p = 0.103) (see Figure. 8). However, there was a significant increase in perceived ease of use between *VR:HHC* and *VR:Eye* (Z = -3.140, p = 0.002).

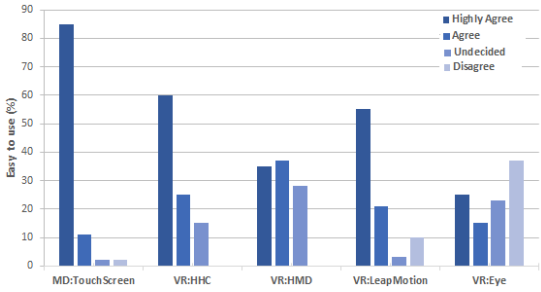


Figure 8: Easy to use: Participants survey on which technique was easier to use. Report show that the mobile device touchscreen was easier to use.

Error Recovery There was a significant difference in error recovery based on interaction technique ($\chi^{2(2)}$ = 12.667, p = 0.002). Significant differences were found between *MD: Touchscreen* and *VR:HMD* as well as *VR:LeapMotion* and *VR:Eye*. In all cases, *MD: Touchscreen* and *VR:HHC* were ranked favorably and there was no significant difference in their ease of error recovery .

6 DISCUSSION

Login Speed Our participants performed significantly better on the mobile device but this was closely matched by their performance while using the *VR:HHC* and *VR:LeapMotion*. We believe that the familiarity of prior usage of mobile devices played a strong part in this outcome and surely with frequent usage of the VR interaction techniques their performances will improve. The data from the quantitative and qualitative (see Figure.7) analysis support our observations.

UsabilityUsability was determined by looking at the quantitative and qualitative data of this study. Our evaluation of the quantitative login speed, error rates and the participant's survey responses about error recovery,speed, likeability and ease of use indicates that the *VR:HHC* and *VR:LeapMotion* was considered as usable as the mobile device *MD:TouchScreen* used in this study (see Figure 9).

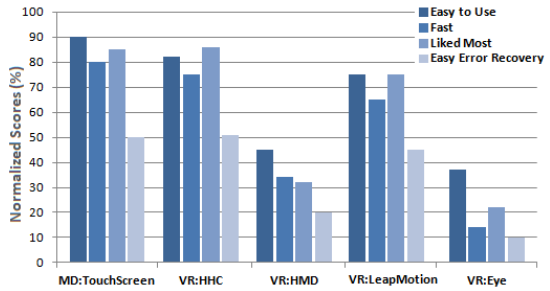


Figure 9: LIKERT Survey: Participants provided subjective feedback in a survey that was analyzed to generate above information .

Shoulder-Surfing Resistance The shoulder-surfing evaluation performed in section 5.1 indicates that SWIPE in VR has a very high resistance to shoulder-surfing (see Figure 6). Thus, making this solution highly secure for public usage where there might be multiple observers, We make this assertion with high confidence because of the fidelity of our experimental design and we are certain that field observations of participants can rarely be as accurate as our experiment.

7 CONCLUSION AND FUTURE WORK

We have presented a detailed and complete study to explore the outcome of porting the popular SWIPE authentication system into the virtual reality environment. Our study used the mobile device version of SWIPE as a control and the reports of the study as stated above, indicates that SWIPE in VR is moderately fast, functionally usable and highly resistant to shoulder-surfing, and this will open doors to the implementation of a familiar and robust SWIPE authentication systems into VR solutions.

Although our study outcomes were satisfactory, we recognize that there is room for improvement due to the short periods of this study. We strongly believe that participants were unable to adequately get familiar with many of the VR interaction techniques that were not inherently intuitive. The above observation might have had some impact on the study outcomes, and we believe there is an opportunity for a future study that fully engages the participants in long periods of VR sessions that requires a series of authentication stages as the session progresses. It is our projection that great improvement will be seen in the participant's login speed, usability, and error-recovery. Lastly, results from the VR shoulder-surfing indicates that the VR implementation of SWIPE has a high resistance, we believe as a future study, it will be interesting to use machine-learning techniques in predicting the participants passwords from the recorded videos of their login activities.

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